MB&P Müller-Boré & Partner
Patentanwälte PartG mbB

Müller-Boré • Friedenheimer Brücke 21 • 80639 München

Europäisches Patentamt
80298 München

November 28, 2024

European Patent Application 22 813 033.2
Applicant: NIKE Innovate C. V.
Our ref.: N05883WOEP - ds/fru

The following is a response to the communication pursuant to Rules 161(1) and 162 EPC dated June 5, 2024:

1. Requests

It is requested to base the grant procedure on the following documents:

- new claims 1 to 15, enclosed as an unmarked version;
- new description pages 1 to 28, enclosed as an unmarked version; and
- figures 1 to 27 (drawing sheets 1/21 to 21/21), as published under the PCT.

If the Examining Division still has objections regarding patentability of the subject-matter of the present application after considering the present response and a subsequent response to a communication

Müller-Boré & Partner
Patentanwälte PartG mbB
Friedenheimer Brücke 21
80639 München
Tel. +49-(0)89/490 57-0
Fax +49-(0)89/450 67 45
Fax +49-(0)89/490 57 10
mbp@mueller-bore.de
www.mueller-bore.de
VAT-No. DE811262789

Patentanwalt I 1, 1a
Rechtsanwalt I 2
European Patent Attorney I 3
European Trademark Attorney I 1
European Design Attorney I 1
Japanese Patent Attorney I 4
Partner gem. PartGG I 5
Dr. W. Müller-Boré (1927-1975)
Dr. Daniele Schiuma, Dipl.-Phys. I 1, 3, 5
Dr. Joachim Hock, Dipl.-Chem. I 1, 3, 5
Dr. Carsten Rocke, Dipl.-Phys. I 1, 3, 5
Christian Haydn, Dipl.-Phys. I 1, 3, 5
Dr. Konstanze Lenhard, Dipl.-Biol. I 1, 3, 5
Dr. Michael Huber, Dipl.-Phys. I 1, 3, 5
Dr. Ulrich Hoffmanns, Dipl.-Chem. I 1, 3, 5
Andreas Ken Schepers, Dipl.-Ing. I 1, 3, 5
Dr. Stephan Beer, M.Sc. (Chem.) I 1, 3, 5
Karsten Schmidt, LL.B., Dipl.-Ing. I 1, 3, 5
Dr. Maria Burger, M.Sc. (Phys.) I 1, 3
Malte Munderloh, Dipl.-Phys. I 1, 3
Dr. Nicolas Streidl, Dipl.-Chem. I 1, 3
Dr. Hans Laub, M.Sc. (Chem.) I 1, 3
Dieter Schaffner, Dipl.-Ing. (FH) I 1, 3
Dr. Robert Knall, Dipl.-Biol. I 1, 3
Elcin Celenk I 2
Yuko Matsuya-Todini, M.Sc. I 3, 4
Alexandra Wendl, Dipl.-Ing. (FH) I 3
Samuel Adams, LL.M., B.S. (Comp.) I 1a, 3
Dr. Markus Schlapps, Dipl.-Phys. I 1, 3
Yoshihito Imai, B.Eng. I 3, 4
Tomoko Kitamura, Dipl.-Inf. I 3
Dr. Reiko Wada-Kröck, M.Sc. (Pharm.) I 3
Dr. Anna Elia, M.Sc. (Phys.) I 1, 3
Hansjörg Neiber, Dipl.-Ing. I 1, 3
Dr. Tilman Hain, Dipl.-Chem. I 1, 3
Francesco Russo, LL.M., M.Sc. (Eng.) I 3
Dr. Helena Schöwerling I 2
Dr. Fabian Hanusch, M.Sc. (Chem.) I 1, 3
Tim Nebas, M.Sc. (Sports Eng.) I 1, 3
Dr. Anne Kunz, Dipl.-Met. I 1, 3
Thomas Hammann, M.Sc. (Aerosp.) I 1, 3
Dr. Christoph Brenninger, M.Sc. I 1, 3
Dr. Nobuhiro Tago, M.Sc. (Life science) I 3
Dr. Johannes Ahner, Dipl.-Chem. I 1
Dr. Dominik Essig, M.Sc. (Chem.) I 1
Andreas Rutetzki (Of Counsel) I 1, 3

Partnerschaftsgesellschaft mbB
Sitz der Gesellschaft: München
Amtsgericht München
Reg.-Nr.: PR 56

Deutsche Bank AG
Promenadeplatz 15
80333 München
BIC DEUTDEDBMUC
IBAN DE16 7007 0024 0271 3220 00

under Article 94(3) EPC, then we would like to make an auxiliary request to schedule a personal interview with an examiner or to schedule oral proceedings in accordance with Article 116 (1) EPC.

2. New Documents

2.1 New claim 1 is based on the combination of claims 1, 4 and 6, as published under the PCT.

2.2 New claims 2 to 8 correspond to claims 2, 3, 5, 7 to 10, as published under the PCT, respectively.

2.3 New claim 9 is based on the combination of claims 11, 14 and 16, as published under the PCT.

2.4 New claims 10 to 14 correspond to claims 12, 13, 15, 17 and 18, as published under the PCT, respectively.

2.5 New claim 15 is based on the combination of claims 19 and 20, as published under the PCT.

2.6 Reference signs have been added to the features of the new claims 1 to 15 in accordance with Rule 43(7) EPC (Rule 6.2(b) PCT).

2.7 New description pages 1 to 28 are based on description pages 1 to 34, as published under the PCT, wherein:

- the relevant prior art document **D1** has been acknowledged and the description has been brought into conformity with the new claims in accordance with Rule 42(1)(b)(c) EPC (Rule 5.1(a)(ii) and (iii) PCT),
- claim like clauses and general statements in the description which imply that the extent of protection may be expanded in some vague and not precisely defined way have been removed in accordance with Art. 84 EPC, and

- the phrase *"incorporated by reference"* has been removed.

With respect to Rule 43(1)(a) EPC (Rule 6.3(b) PCT), it is noted that the two-part form has not been chosen for new independent claims 1 and 9, since the two-part form would give a distorted and misleading picture of the subject-matter of new claim 1 and 9 and the prior art. Accordingly, in the present case, the two-part-form is unsuitable referring to the Guidelines for Examination, Part F, Chapter IV, 2.3 "Two-part form unsuitable".

3. Conciseness (Article 84 EPC)

According to the ISA/EP, independent claims 1 and 11 appear to relate to the same subject matter and to differ from each other only regarding the definition of the subject matter for which protection is sought and/or in respect of the terminology used for the features of that subject matter (see item 4.1 of the WO/ISA). Consequently, according to the ISA/EP, claims 1 and 11 lack conciseness within the meaning of Article 6 PCT.

The Applicant respectfully disagrees. In fact, claims 1 and 11 comply with the requirements of Article 84 and Rule 43(2) EPC. Particularly, independent claims 1 and 11 relate to alternative solutions to a particular problem, where it is inappropriate to cover these alternatives by a single independent claim. Accordingly, independent claims 1 and 11 are concise and both can be maintained in the set of claims without violating the requirements of Article 84 EPC.

As above indicated, claims 1 and 11 of the present application define alternative solutions to a particular problem. Particularly, the claimed invention solves the technical problem of improving the cushioning and energy distribution provided by the sole structure (see section 6 below).

A first solution to this problem is provided by claim 1. Claim 1 relates to a sole structure for an article of footwear. The sole structure comprises a first cushion extending continuously from an anterior end of the sole structure to a posterior end of the sole structure and including an upper side, a bottom side, formed on an opposite side from the upper side, and a recess formed in the upper side. The sole structure also comprises

a first bladder disposed within the recess adjacent to a lateral side of the sole structure and a second bladder disposed within the recess adjacent to a medial side of the sole structure. The recess extends from a first end surface adjacent to a toe portion of the sole structure to a second end surface disposed in one of a forefoot region or a mid-foot region of the sole structure. Advantageously, at least one of the first end surface and the second end surface is oriented at an oblique angle relative to a central axis of the first bladder. This improves the cushioning and energy distribution provided by the sole structure.

A second solution to the technical problem is provided by claim 11. Claim 11 relates to a sole structure for an article of footwear. The sole structure comprises a cushion extending continuously from an anterior end of the sole structure to a posterior end of the sole structure and including a top side, a bottom side formed on an opposite side from the top side, and a receptacle disposed between the top side and the bottom side. A first bladder is disposed within the receptacle adjacent to a lateral side of the sole structure, and a second bladder is disposed within the receptacle adjacent to a medial side of the sole structure. The receptacle extends from a first end surface adjacent to a toe portion of the sole structure to a second end surface disposed in one of a forefoot region or a mid-foot region of the sole structure. Advantageously, at least one of the first end surface and the second end surface is oriented at an oblique angle relative to a central axis of the first bladder. This configuration improves the cushioning and energy distribution provided by the sole structure.

As evident from the above, the claimed solutions solve the addressed technical problem by some common features, that is a first bladder, a second bladder, and the arrangement of the first and second end surfaces. However, the claimed solutions also comprise different and mutually exclusive features, that is claim 1 is directed to a sole structure including a cushion formed by a plurality of portions or subcomponents (first cushion and second cushion), whereas claim 11 is directed to a sole structure including a cushioning element formed as a monolithic structure.

Therefore, the sole structure as defined in claims 1 and 11 are structurally different and not equivalent. Due to the alternative and mutually exclusive features of original claims

1 and 11, these claims provide alternative solutions, and it would not be appropriate to cover these embodiments by a single independent claim.

It is therefore considered that independent claims 1 and 11 (new claims 1 and 9) are concise and, thus, comply with the requirements of Article 84 EPC.

4. Prior Art

Document **D1** ("US 5 313 717") describes a reactive energy fluid filled cushioning and stabilizing apparatus for a shoe comprising one or more anatomically shaped fluid filled bladders that are positioned in the forefoot, arch and/or heel areas of the shoe. The bladders assume a complementary custom fitting configuration to the contours of a foot inserted in the shoe. The fluid filled bladders are positioned separate from each other in the forefoot, arch and heel areas of the shoe and are interconnected in fluid communication by fluid conducting channels, or the bladders are positioned independent of each other in the forefoot, arch and heel areas of the shoe.

Document **D2** ("US 5 771 606") describes a support and cushioning system for an article of footwear. The system includes a resilient insert disposed between a midsole and an outsole of a shoe. The resilient insert includes several chambers disposed in a heel portion of the resilient insert. These chambers are fluidly interconnected to each other via periphery passages. The resilient insert also includes several chambers disposed in a forefoot portion of the resilient insert. These chambers are also fluidly interconnected to each other. A connecting passage connects the chambers in the heel portion and the chambers in the forefoot portion of the resilient insert. A bladder having a fluidly interconnected heel chamber and forefoot chamber is also inserted above the midsole to provide added cushioning to the wearer.

Document **D3** ("US 2020/305551 A1") describes a sole structure for an article of footwear including a bladder and a cushioning member. The bladder includes a first fluid-filled segment including first and second terminal ends disposed on opposite ends of the first fluid-filled segment along a first longitudinal axis of the first fluid-filled segment. The first longitudinal axis extends in a first direction from a heel region of the article of

footwear to a forefoot region of the article of footwear. A second fluid-filled segment is spaced apart from the first fluid-filled segment in a second direction transverse to the first direction and includes third and fourth terminal ends disposed on opposite ends of the second fluid-filled segment along a second longitudinal axis of the second fluid-filled segment. A third fluid-filled segment extends in the second direction between the first fluid-filled segment and the second fluid-filled segment. The bladder is received in the cushioning member.

Document **D4** ("US 2003/208930 A1") describes a sole component for footwear. The chamber can be formed as a single bladder chamber in contact with an elastomeric midsole, or a single chamber formed by sealing a void in elastomeric material. The interface between the chamber and elastomeric material is sloped and gradual so that the shape of the chamber and its placement in a midsole determine the combination of response characteristics in the sole component. The chamber has a relatively simple shape with one axis of symmetry with a rounded portion and a narrow portion. Varying the placement of the chamber in the elastomeric material can simulate the impact response of more complex and expensive systems with only a single chamber shape that needs to be stocked. The chamber has a relatively large volume, is devoid of internal connections, and has an internal pressure within 5 psi of ambient pressure, and preferably at ambient pressure. Since air is used as the fluid, no specialized gases are required. No specialized films or bladder materials are required where the chamber is formed as a bladder, since the bladder is not highly pressurized.

5. Novelty - Art. 54(1) and (2) EPC - Art. 33(2) PCT

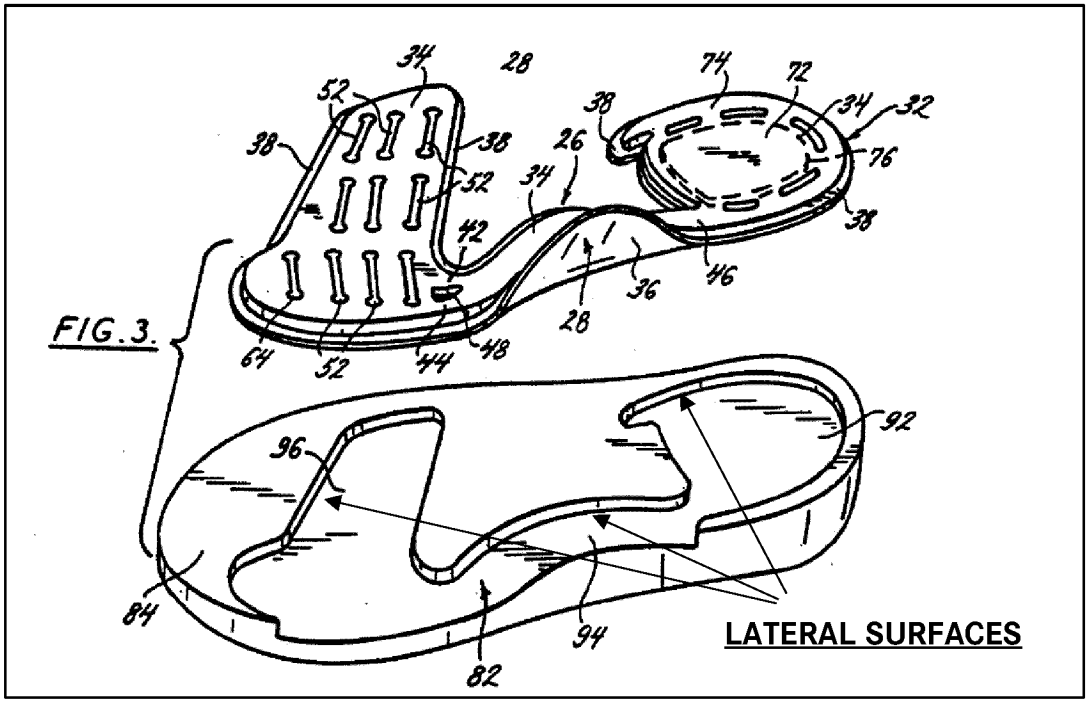
The subject matter of new **claim 1** is novel within the meaning of Article 54(1) EPC.

Particularly, none of the cited references **D1** to **D4** discloses at least the following combination of features, as defined in new claim 1 of the present application:

> *"at least one of the first end surface (160) and the second end surface (162) is oriented at an oblique angle relative to a central axis (A_{108}) of the first bladder (108).".*

According to the ISA/EP, the subject matter of dependent claim 6 is allegedly anticipated by **D1** (see item 2.3 of the WO/ISA). Particularly, this objection is based on **Fig. 3** of **D1**. The Applicant respectfully disagrees. In fact, the subject matter of claim 6 is not disclosed in a direct and unambiguous manner by **D1**.

D1 describes an apparatus (20) for a shoe (22) comprising one or more anatomically shaped fluid filled bladders (24, 26, 28, 32) that are positioned in the forefoot, arch and/or heel areas of the shoe (see Fig. 3 of **D1** below). To position the apparatus (20) in the sole of the shoe (22), a recessed cavity (82) is formed in a top surface of the shoe sole (84). As shown in Fig. 3 of **D1**, the recessed cavity (82) has an outline configuration substantially identical to the outline configuration of the apparatus (20).



According to the ISA/EP, the features of dependent claim 6 are allegedly disclosed by Fig. 3 of **D1** (see item 2.3 of the WO/ISA). However, as evident from Fig. 3 of D1, the lateral surfaces of the recessed cavity (82) are clearly **vertical** with respect to the bottom surface of the cavity (82). It is further noted that **D1** is completely silent about the arrangement of the lateral surfaces of the recessed cavity (82) with respect to the bladders and/or with respect to the bottom surface of the recessed cavity (82).

Therefore, the subject matter of claim 6, according to which at least one of the first end surface (160) and the second end surface (162) is oriented at an oblique angle relative to a central axis (A_{108}) of the first bladder (108), cannot be disclosed by **D1**.

The subject matter of new **claim 1** (combination of claims 1, 4 and 6 as published under the PCT) is also novel over the cited documents **D2** to **D4**. In this respect, it is noted that documents **D2** to **D4** are all marked as **"A"** documents in connection with the subject matter of dependent claims 2 to 10 and 12 to 20 (see the Form PCT/ISA/210 enclosed to the ISR).

Therefore, the subject-matter of new claim 1 and, *mutatis mutandis,* of new claim 9 has the required novelty, as required by Article 54(1) EPC.

6. Inventive Step - Art. 56 EPC - Art. 33(3) PCT

The subject-matter of new claim 1 also involves the necessary inventive step:

6.1 Closest prior art

When applying the problem and solution approach, document **D1** is the closest prior art with respect to the subject-matter of new claim 1.

6.2 Differentiating features

The subject-matter of new claim 1 differs from that of document **D1** in that:

> *"at least one of the first end surface (160) and the second end surface (162) is oriented at an oblique angle relative to a central axis (A_{108}) of the first bladder (108).".*

6.3 Technical effect

The technical effect achieved by the differentiating features between the subject-matter

of new claim 1 and that of document **D1** is that an expansion space having a trapezoidal cross-sectional profile is provided between the bladders and the cushioning member. As a result, when the sole structure is compressed, the bladders may laterally deform and extend into the expansion space without contacting each other or the end surfaces (see, e.g., paragraphs [0059] to [0061] of the description as originally filed). Accordingly, the cushioning and energy distribution provided by the sole structure are improved.

6.4 Objective-technical problem

Thus, starting from document **D1** as the closest prior art and in light of the features not disclosed by document **D1**, the skilled person is confronted with the objective technical problem of

improving the cushioning and energy distribution provided by the sole structure.

6.5 Non-obvious solution

The skilled person starting from document **D1** and attempting to solve this problem does not obtain any suggestions from document **D1** leading toward the subject-matter of new claim 1.

Document **D1** describes an apparatus (20) for a shoe (22) comprising one or more anatomically shaped fluid filled bladders (24, 26, 28, 32) that are positioned in the forefoot, arch and/or heel areas of the shoe (see Fig. 3 of **D1**). A recessed cavity (82) is formed in a top surface of the shoe sole (84) to contain the fluid filled bladders. As shown in Fig. 3 of **D1**, the recessed cavity (82) is configured to have an outline configuration substantially identical to the outline configuration of the apparatus (20).

However, document **D1** neither teaches nor suggests the features outlined in section 6.2 above.

In fact, when starting from document **D1**, to address the above technical problem, the skilled person would rather consider modifying the cross-sectional areas of the channels extending between the forefoot, arch and heel bladders to modify and improve the cushioning and energy distribution provided by the sole structure (see Col. 3, lines 23-42 of D1).

However, document **D1** does not provide any hints nor suggestions regarding the distinguishing features of new claim 1.

In fact, at no point in **D1**, the skilled person is motivated to modify the arrangement of the lateral surface of the recessed cavity (82) of **D1** to be oriented at an oblique angle relative to a central axis of the bladder, as required by new **claim 1**.

Thus, document **D1** cannot render the subject-matter of new claim 1 obvious.

Even when further considering any of documents **D2** to **D4**, the skilled person would not have arrived at the subject-matter of new claim 1 in an obvious manner. Documents D2 to D4 neither teach nor suggest at least one of the first end surface and the second end surface oriented at an oblique angle relative to a central axis of the first bladder, as set forth in new claim 1.

For these reasons, the subject-matter of new claim 1 of the present application is not obvious for the skilled person with respect to any of documents **D1** to **D4**, neither taken alone nor combined with each other.

It is stressed that when assessing inventive step the question to be answered (see EPO Guidelines G-VII, 5.3) is whether there is any teaching in the prior art as a whole that **would** (not simply could, but would) have prompted the skilled person, faced with the objective technical problem, to modify or adapt the closest prior art while taking account of that teaching, thereby arriving at something falling within the terms of the claims, and thus achieving what the invention achieves. In other words, the point is not whether persons skilled in the art could have arrived at the invention by adapting or modifying the

closest prior art, but whether they would have done so because the prior art incited them to do so in the expectation of some improvement or advantage, which in view of the above presently is not the case.

Therefore, the subject-matter of new claim 1 and, *mutatis mutandis,* of new claim 9 involves the necessary inventive step, as required by Article 56 EPC.

Accordingly, the above requests are **well founded.**



Christian Haydn
European Patent Attorney
(in absence of Dr. Daniele Schiuma)

Encls.:
- New claims 1 to 15 (marked-up and unmarked versions)
- New description pages 1 to 28 (marked-up and unmarked versions)